Two remarks on Bedert's "On unique sums in Abelian groups"

Mark Watson

21 July 2026

This note records two observations about Bedert's paper [B]. First, the upper bound $m(p) = O((\log p)^2)$ of [B, Theorem 5] already follows, with a worse constant, from a theorem of Łuczak and Schoen [LS], which is [B]'s own reference [9] (Section 1). Second, [B, Proposition 1] holds with its first binomial factor removed (Section 2); this is the one result proved in full here, and it is machine-checked in Lean, apart from its final, elementary Vandermonde step. Section 3 collects smaller observations from a close reading of [B]'s proofs, about information that the packaged statements discard, together with an elementary limiting construction and some computations. Everything else is context, and is [B]'s or [LS]'s.

For $A \subseteq \mathbb{Z}/p\mathbb{Z}$ and $g \in \mathbb{Z}/p\mathbb{Z}$, write

$$r_A(g) \;=\; |\{(a, a') \in A^2 : a + a' = g\}|,$$

counting ordered pairs. A has *no unique sum* iff no g satisfies $1 \le r_A(g) \le 2$. Following [B], $m(p)$ is the size of a smallest $A \subseteq \mathbb{Z}/p\mathbb{Z}$ with no unique sum; I take $|A| \ge 2$ throughout, which excludes the vacuous $A = \emptyset$.

1 $m(p) = O((\log p)^2)$ follows from [LS, Theorem 2] with $Q = 2$

[LS] work with the same ordered count as r_A. They set

$$\nu(t) \;=\; \nu_A(t) \;=\; |\{(a, b) \in A \times A : t = a + b\}|, \qquad \nu(A) \;=\; \min_{t \in A+A} \nu(t),$$

with the convention (their p. 101) that "if $a \neq b$, then we view $t = a + b$ and $t = b + a$ as two different representations of t". Their Theorem 2 (p. 109) states:

Theorem 2 ([LS], p. 109)**.** *"For every positive integer $Q < \log_2 p/(2\log_2(\gamma_p \log_2 p))$, where γ_p is the constant given in Straus' construction, there exists a set $A \subseteq \mathbb{Z}/p\mathbb{Z}$ such that $|A| = (\gamma_p \log_2 p)^Q$ and $\nu(A) \geq 2^Q$."*

They record $\gamma_p \leq 2$, tending to $2/\log_2 3$.

Take $Q = 2$. The hypothesis reads

$$\log_2 p \ > \ 4\log_2(\gamma_p \log_2 p),$$

which holds for $p \geq 2^{22}$. The resulting A has

$$|A| \ \leq \ 4(\log_2 p)^2, \qquad \nu(A) \ \geq \ 4,$$

the first bound using $\gamma_p \leq 2$. Every g with $r_A(g) \geq 1$ lies in $A + A$, so $r_A(g) = \nu_A(g) \geq 4 > 2$: no g has $r_A(g) \in \{1, 2\}$, and A has no unique sum. Hence

$$m(p) \ \leq \ 4(\log_2 p)^2 \qquad (p \geq 2^{22}).$$

This is the bound of [B, Theorem 5]. [LS] never state the $m(p)$ corollary, though the step from $\nu \geq 4$ to "no unique sum" is immediate, and [B] (p. 3) records the previous best upper bound as "$m(p) \ll \sqrt{p}$ which came from a rather easy construction".

Bedert's construction is a constant-factor improvement over the one above, using balanced sets in place of Straus's S. At $Q = 2$ the [LS] set is $S + x \cdot S$, where [LS, Lemma 4] supplies an x making $(s, s') \mapsto s + xs'$ injective, so $|S + xS| = |S|^2$: the same two-dimensional shape as Bedert's $B \times B$, though [LS, Lemma 4] gives injectivity only, not the higher-order condition a Freiman isomorphism requires.

2 A redundant factor in [B, Proposition 1]

Let $Z = (z_1, \ldots, z_n)$ be a finite indexed family in an abelian group, let

$$\Sigma(Z) \ = \ \left\{ \sum_{i \in I} z_i \ : \ I \subseteq [n] \right\}$$

be its set of subset sums, and let $d = \dim(Z)$ be the largest size of an index set whose subset sums are pairwise distinct (equivalently, indexing a dissociated subfamily). The trivial bound $|\Sigma(Z)| \leq 2^n$ always holds, with equality when Z itself is dissociated, and [B, Proposition 1] shows this is essentially the only way $\Sigma(Z)$ gets large: a big additive span forces a big dissociated subfamily. Quantitatively, it states

$$|\Sigma(Z)| \ \leq \ \binom{n}{d}\binom{n+d}{d}.$$

The observation here is that the first factor can be dropped.

Proposition.

$$|\Sigma(Z)| \;\le\; \#\{I \subseteq [n] : I \text{ dissociated}\} \;\le\; \sum_{j=0}^{d} \binom{n}{j} \;\le\; \binom{n+d}{d}. \qquad (1)$$

Proof. The idea is to pick one canonical index set for each attainable sum, and to count the representatives by shattering. Order index sets by their characteristic 0–1 strings, lexicographically. For each $y \in \Sigma(Z)$ let R_y be the lex-least I with $\sum_{i \in I} z_i = y$, and set

$$F \;=\; \{R_y : y \in \Sigma(Z)\},$$

so that $|F| = |\Sigma(Z)|$.

Every set S shattered by F is dissociated. Otherwise there are distinct $P, Q \subseteq S$ with equal sums; cancelling $P \cap Q$ we may take them disjoint, say $P < Q$ lexicographically. Shattering gives $R \in F$ with $R \cap S = Q$; then

$$R' \;=\; (R \setminus Q) \cup P$$

has the same sum as R and is lex-smaller, contradicting the minimality of R.

By Pajor's strengthening [P] of the Sauer–Shelah lemma [S, Sh], a finite family shatters at least as many sets as it has members, so F shatters at least $|F|$ sets, all dissociated. A dissociated index set has at most d elements by definition of dim, and

$$\sum_{j \le d} \binom{n}{j} \;\le\; \binom{n+d}{d}$$

by Vandermonde. $\square$

The argument uses only the indexed structure, so it covers multisets, the setting of [B].

Bedert's own extremal example ([B] p. 6: Z consists of k copies of each of the standard basis vectors $e_1, \ldots, e_d$ of $\mathbb{Z}^d$, $n = kd$) does not obstruct the improvement: there the first inequality of (1) is an equality, both sides being $(k+1)^d$, since a dissociated index set picks at most one of the k copies of each e_i.

Downstream, with $K = n/d$, [B, Corollary 1] becomes

$$|\Sigma(Z)| \;\le\; \binom{n+d}{d} \;\le\; \big(e(K+1)\big)^d,$$

of K^d scale rather than the K^{2d} of the stated $(4K)^{2d}$.

Remark. The first two inequalities of (1) are machine-checked in Lean, as `BedertLab.card_subsetSums_le` in the repository accompanying this note, with Mathlib's `Finset.card_le_card_shatterer` as the classical input (the Lean proof uses the colexicographic order; the argument is the same). The final Vandermonde step is not formalized; `verify.py` checks it numerically.

3 Further observations

These are secondary, none of them is asserted as new, and a remark after each one states exactly what backs it.

The first three concern the lower-bound machinery of [B], so a word of context. That proof runs an iteration (Propositions 5 and 6) in which a set S_j of shifts is repeatedly enlarged while $D + S_j$ covers more of A, for a fixed dissociated $D \subseteq A$. Coverage grows by at least $|D|^2/(36|A|)$ per step, so the iteration stops within $36K^2$ steps, where $K = |A|/\dim(A)$; and it can only stop once $|S_j|$ has outgrown the constraint [B, (22)]. A lower bound on the number of steps is therefore a lower bound on K. [B] writes ω_1 for the resulting bound: applying Proposition 5 (to a translate of A, as in the proof of Theorem 4) gives $K \geq \omega_1(p)$, with $\omega_1(p) \gg (\log\log\log p)^{1/2}$, and the conversion in the proof of his Theorem 4 turns this into the $\omega(p)$ of the main result.

3.1 The shift set in Proposition 6 grows polynomially

[B, (23)] records only $|S'| \leq \max(2|S|, |S|^3)$, so the iteration in Proposition 5 knows only $\log_2 |S_j| \leq 3^j$. Reading the proof, however, S' always has one of two forms:

$$S' \;=\; S \cup (S+t) \;\text{ for a single } t, \qquad \text{or} \qquad S' \;=\; 2S - S \quad (\text{using } 0 \in S);$$

these are the "first case" and "final case" of the proof, pp. 24–25.

Read as generalized arithmetic progressions, the two operations are "append a side of length 2" and "map every side L to $3L-2$", and tracking the sides through the iteration gives

$$\log_2 |S_j| \;\leq\; 0.4\, j^2 + j, \tag{2}$$

polynomial rather than exponential in j. With the same termination argument this yields

$$\omega_1(p) \;\gg\; (\log\log p)^{1/4} \qquad \text{in place of} \qquad (\log\log\log p)^{1/2}.$$

In words: with $|S_j|$ growing only polynomially, far more steps are needed before [B, (22)] fails, and every extra step raises the lower bound on K.

The tracked progression is never a hypothesis, only an observation about the sets the proof already constructs, so no lemma in Proposition 6 is affected.

Remark. `verify.py` checks (2) for all 2^j schedules with $j \leq 14$. A Lean proof for arbitrary schedules, and a kernel-checked derivation of the exponent from (2), exist in my working development but are not included in the repository; the passage from the two-shape reading of Proposition 6 to (2) is on paper only. The two-shape claim and its exhaustiveness were read from the PDF.

3.2 The two branches of Proposition 6 have different increments

[B, (24)] records an increment of $|D|^2/(36|A|)$ for both branches. Inspection of the proof, however, gives

$$\frac{|D|^2}{36|A|} \text{ for the translate branch,} \qquad \frac{|D|^2}{12|A|} \text{ for the cube branch,}$$

three times larger, and (24) takes the minimum; nothing downstream restores the difference, since Proposition 5 consumes only the packaged statement. With c translate and f cube steps the termination budget is therefore

$$c + 3f \ \leq \ 36K^2 \qquad \text{rather than} \qquad c + f \ \leq \ 36K^2,$$

which divides the quadratic coefficient in the growth analysis by 3 and multiplies the final constant by exactly $3^{1/4} \approx 1.316$.

Remark. Read from the PDF's displayed formulas by two independent passes reaching identical constants; not formalized.

3.3 What the ω_1 improvement is worth

The gain to ω_1 is immediate and large; the gain to the published ω is not. [B]'s conversion

$$f(x) \ = \ \frac{x}{2(2 + \log_2 x)}$$

penalizes a larger ω_1 more, because its denominator grows with x. At $\log_2 \log_2 p = 2^{64}$, [B]'s $\omega_1 \approx 1.06$ loses a factor 4.2 under f, while $\omega_1 \approx 1.4 \times 10^4$ loses a factor 31.5.

Comparing the published constant-free shapes,

$$\frac{\sqrt{\log\log\log p}}{\log\log\log\log p} \qquad \text{against} \qquad \frac{(\log\log p)^{1/4}}{\log\log\log p},$$

the crossover past which the improved ω exceeds [B]'s is around $\log\log p \approx 5\times 10^4$, that is $p \approx \exp(\exp(50000))$. So this is an asymptotic improvement to ω and an immediate one only to ω_1.

Remark. The numbers in this subsection are recomputed by `verify.py`.

3.4 A limit on the Section 2 improvement

The Proposition of Section 2 removes one factor; the remaining K^d scale cannot be removed, even for sets. Take $d = 2h$ and

$$Z \;=\; \{\, e_i + \chi_S \;:\; 1 \le i \le h,\ S \subseteq \{h+1,\dots,2h\}\,\} \;\subset\; \mathbb{Z}^d,$$

a set, not a multiset. Its subset sums record, in the first h coordinates, how many elements with each index were chosen, giving

$$\log_2 |\Sigma(Z)| \;\ge\; d^2/4.$$

But any dissociated $D \subseteq \{0,1\}^d$ has $2^{|D|}$ distinct subset sums lying in $\{0,\dots,|D|\}^d$, so

$$2^{|D|} \;\le\; (|D|+1)^d,$$

and $\dim(Z) = O(d\log d)$. The ratio diverges, so no bound of the form $|\Sigma(Z)| \le 2^{C\dim(Z)}$ holds for sets.

Remark. Elementary; both halves are reproduced in `verify.py`.

3.5 Exact values of $m(p)$

Computed by exact solver, certified optimal:

p	3	5	7	11	13	17	19	23	29	31	37
$m(p)$	3	4	5	7	7	8	9	10	11	11	12

with $m(41), m(43) \le 13$. I did not find this sequence in the OEIS.

The solver's optimal witnesses for $p \le 23$ happen to be double-covering additive bases, though at $p = 19$ and 23 most extremal sets are not additive bases at all; $m(p)$ tracks $2\sqrt{p}$ throughout this range. That is a warning rather than a finding: the $2\sqrt{p}$ design regime governs until roughly $p \sim 10^3$

to 10^4, so these values cannot identify the asymptotic exponent, and fitting a growth law to $p < 1000$ fits the wrong regime.

Separately, the minimum size of a balanced set is $\lceil \log_2 p + 1 \rceil + \{0, 1\}$ throughout the computed range, so the Bilu–Lev–Ruzsa rectification threshold $\log_2 p$ ([B, Lemma 1]) is essentially sharp here.

Remark. The values of $m(p)$ for $p \leq 19$, and the balanced-set minima on the same range, were reproduced by independent exhaustive search, which `verify.py` repeats.

Closing remarks

Section 2 is a direct application of a classical theorem (Sauer [S], Shelah [Sh], Pajor [P]), and while I could not locate it stated as a subset-sum bound, that is the profile of a fact a specialist may regard as standard. I am not affiliated with an institution, so MathSciNet and some publisher citation indices were unavailable to me. I made substantial use of AI tools in this work; the quotations were checked character for character against the sources, and the Proposition of Section 2 is machine-checked in Lean, apart from the final Vandermonde step.

To check, in the repository (`github.com/mkwatson/unique-sums-notes`): run `lake exe cache get && lake build`, then

`#print axioms BedertLab.card_subsetSums_le`,

expecting `[propext, Classical.choice, Quot.sound]`; and run `python3 verify.py` for the arithmetic and the small computations.

References

[B] B. Bedert, *On unique sums in Abelian groups*, Combinatorica **44**(2) (2024), 269–298; arXiv:2303.15134. Page references are to the arXiv PDF.

[LS] T. Łuczak and T. Schoen, *On a problem of Konyagin*, Acta Arith. **134**(2) (2008), 101–109.

[P] A. Pajor, *Sous-espaces l_1^n des espaces de Banach*, Travaux en Cours 16, Hermann, Paris, 1985.

[S] N. Sauer, *On the density of families of sets*, J. Combin. Theory Ser. A **13** (1972), 145–147.

[Sh] S. Shelah, *A combinatorial problem; stability and order for models and theories in infinitary languages*, Pacific J. Math. **41** (1972), 247–261.